

Validation Architecture in delaunay

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Author TODO. Write the abstract in your own words.

Additional Key Words and Phrases: TODO author-supplied keywords

1 INTRODUCTION

- **Author TODO.** State the problem and motivation.
- **Author TODO.** Explain why validation architecture matters for a scientific computational-geometry library.
- **Author TODO.** Summarize the central contribution in your own words.
- **Author TODO.** Preview the paper structure.

2 BACKGROUND AND DEFINITIONS

- **Author TODO.** Define the triangulation, simplex, ridge, facet, and realization terms used throughout the paper.
- **Author TODO.** Introduce PL-manifold terminology and the link condition.
- **Author TODO.** Introduce Euclidean, toroidal, and spherical realization models.
- **Author TODO.** Introduce Delaunay and future geometric-predicate terminology.

3 VALIDATION HIERARCHY

3.1 Level 1: Element Validity

- **Author TODO.** Describe what individual vertices, simplices, facets, and coordinates must prove locally.
- **Author TODO.** Explain what this level deliberately does not check.

3.2 Level 2: Combinatorial Consistency

- **Author TODO.** Describe adjacency, incidence, neighbor symmetry, and TDS integrity.
- **Author TODO.** Distinguish coherent stored simplex orderings, including periodic quotient facets, from intrinsic orientability.
- **Author TODO.** Explain the separation from topology and geometry.

3.3 Level 3: Intrinsic PL Topology

- **Author TODO.** State the PL-manifold and boundary conditions.
- **Author TODO.** Discuss pseudomanifold and strict PL-manifold policy choices.
- **Author TODO.** Describe the 2D/3D intrinsic orientation witness and periodic quotient parity constraints.
- **Author TODO.** Explain why this layer is independent of Euclidean, toroidal, and spherical realizations.

3.4 Level 4: Valid Realization

- **Author TODO.** Use the terminology stack: Level 4 Valid Realization; Euclidean/toroidal valid affine-chart realization; spherical valid spherical realization; Level 5 Geometric Predicate Satisfaction / Delaunay Optimality.
- **Author TODO.** Describe valid realization in the active ambient model.

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delaunay validation architecture

Level 1 checks local objects; Levels 2-3 are intrinsic; Level 4 checks valid realization; Level 5 checks geometric predicates.

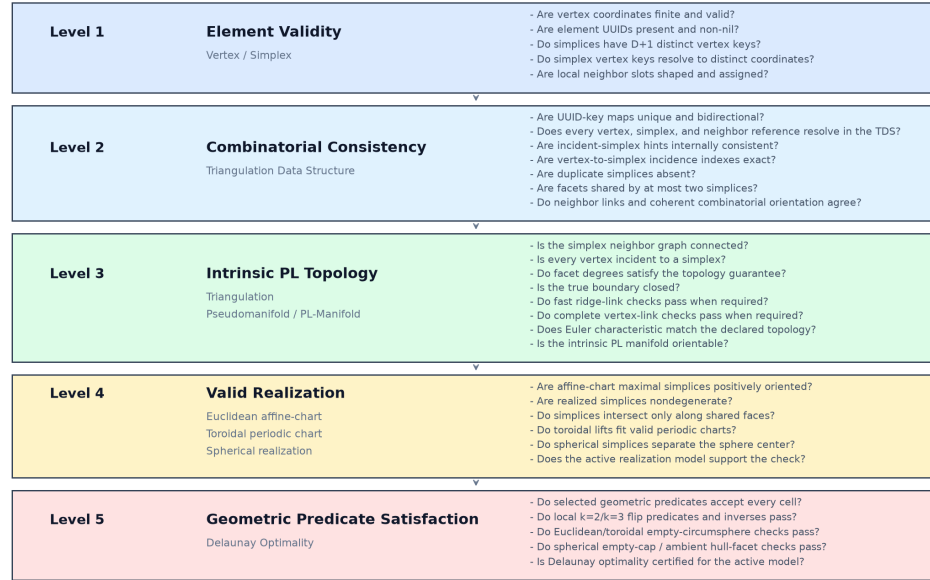


Fig. 1. Overview of the validation hierarchy in delaunay

- **Author TODO.** Compare Euclidean/toroidal affine-chart realizations and spherical realizations.
- **Author TODO.** Keep positive geometric orientation distinct from Level 2 stored coherence and Level 3 intrinsic orientability.
- **Author TODO.** Explain why realization validity is a precondition for geometric predicates.

3.5 Level 5: Geometric Predicates

- **Author TODO.** Describe Delaunay as the first implemented predicate family.
- **Author TODO.** Explain how Euclidean, toroidal, and spherical Delaunay predicates differ.
- **Author TODO.** Leave room for regular, weighted, constrained, Gabriel, alpha, and future predicate families.

4 IMPLEMENTATION ARCHITECTURE

Table 1. Public validation API by owning layer.

Level	Canonical public surface
1	<code>Vertex::is_valid / vertex_report</code> ; <code>Simplex::is_valid / simplex_report</code>
2	<code>Tds::is_valid / structure_report</code> ; <code>Tds::validate</code> certifies Levels 1–2
3	<code>Triangulation::is_valid_topology / topology_report / orientation_witness</code> ; <code>Triangulation::validate</code> certifies Levels 1–3
4	<code>Triangulation::is_valid_realization / realization_report</code> ; <code>validate_realization</code> certifies Levels 1–4
5	<code>DelaunayTriangulation::is_valid_delaunay / delaunay_report</code> ; <code>DelaunayTriangulation::validate</code> certifies Levels 1–5

- **Author TODO.** Map the five validation levels onto the Rust API surface.
- **Author TODO.** Describe the role of fast-fail checks, diagnostics, reports, and cumulative validation.
- **Author TODO.** Explain construction-time validation policy choices such as `ExplicitOnly` and `OnSuspicion`.
- **Author TODO.** Describe how spherical topology fits as a coordinate/metric backend rather than a new simplex dimension.

5 DIAGNOSTICS AND REPRODUCIBLE FIGURES

Level 1 — Element Validity

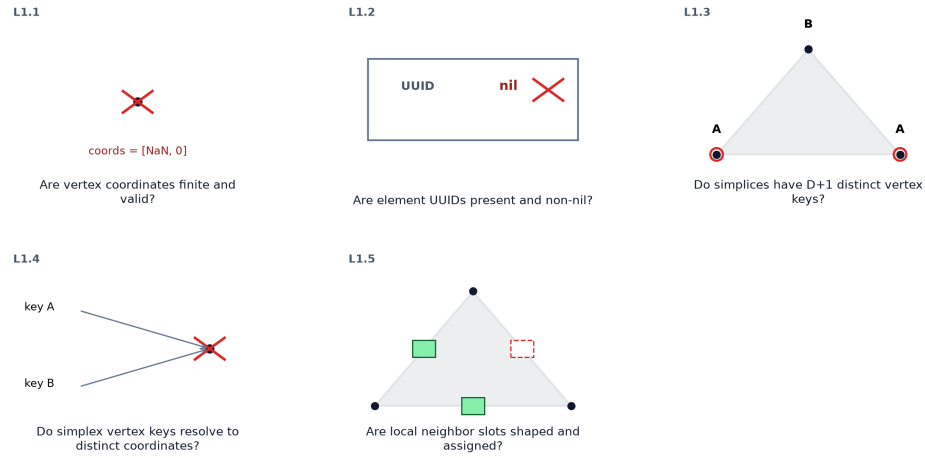


Fig. 2. **Author TODO.** Describe the Level 1 validation witness figure.

Level 2 — Combinatorial Consistency

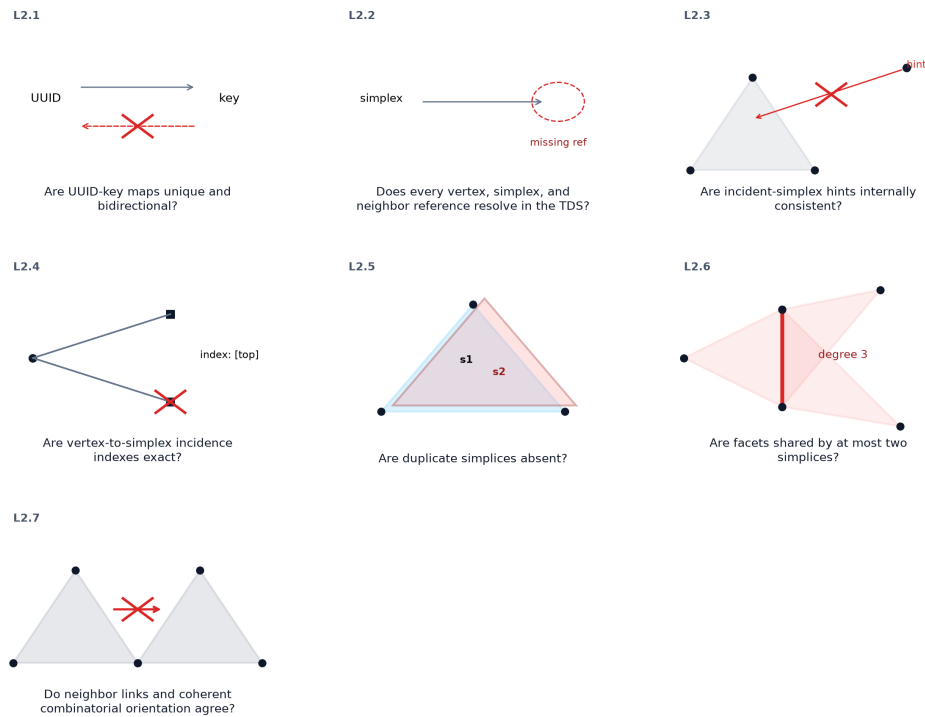


Fig. 3. **Author TODO.** Describe the Level 2 validation witness figure.

Level 3 — Intrinsic PL Topology

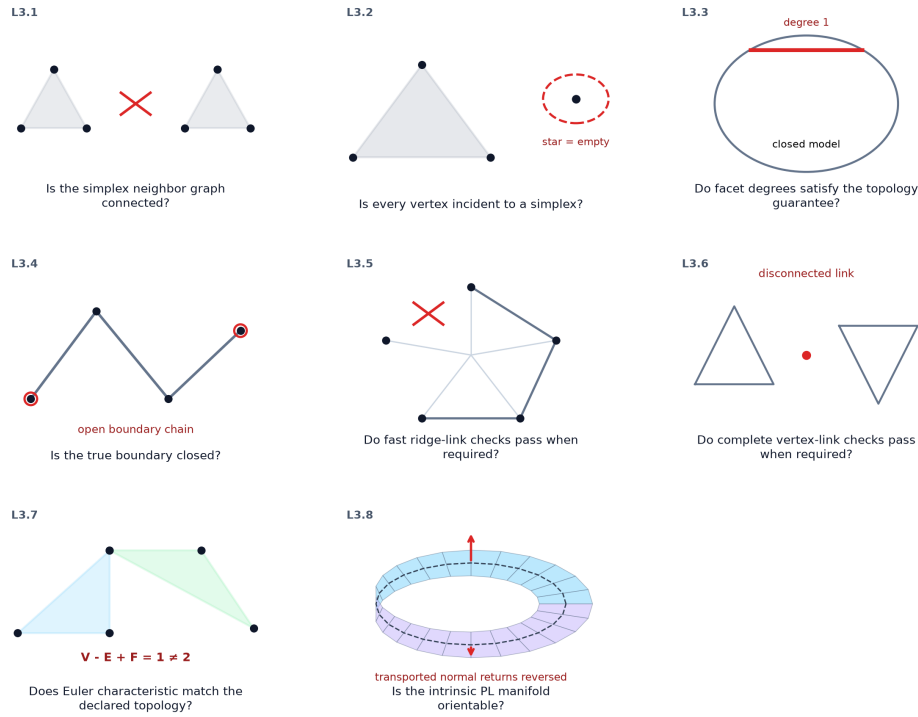


Fig. 4. **Author TODO.** Describe the Level 3 validation witness figure.

Level 4 — Valid Realization

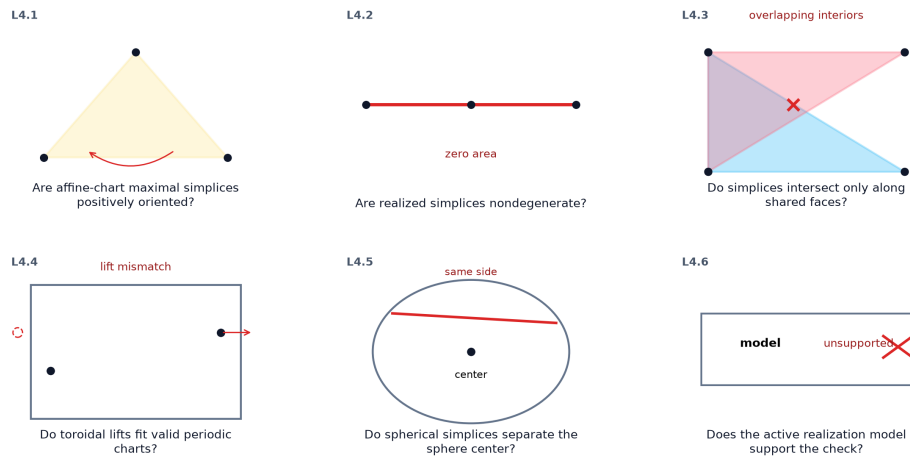
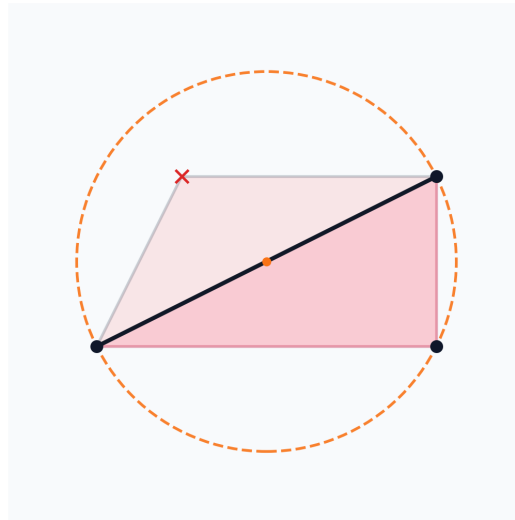


Fig. 5. **Author TODO.** Describe the Level 4 validation witness figure.



Do Euclidean/toroidal
empty-circumsphere checks pass?

Fig. 6. **Author TODO.** Describe the Level 5 validation witness figure.

- **Author TODO.** Explain how the validation notebook generates paper figures.
- **Author TODO.** Describe representative diagnostics without duplicating API documentation.
- **Author TODO.** Connect figure generation to the reproducibility story.

6 RELATED WORK

- **Author TODO.** Place PL topology references and Pachner/Lickorish results.
- **Author TODO.** Place Euclidean robust-predicate and Delaunay references.
- **Author TODO.** Place spherical Delaunay and Riemannian-manifold references.
- **Author TODO.** Clarify which claims are architectural observations rather than literature claims.

7 DISCUSSION AND FUTURE WORK

- **Author TODO.** Discuss extensibility to additional realization models and predicate families.
- **Author TODO.** Discuss limitations of the current implementation.
- **Author TODO.** Identify open validation and benchmarking questions.

8 CONCLUSION

- **Author TODO.** Write the conclusion in your own words.

ACKNOWLEDGMENTS

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